MASTER FUNCTIONAL SKILLS (MFS) · WORKED SOLUTIONS ·
MasterFunctionalSkills.co.uk · AQA Maths Level 1 Specialist
Full Marks (100% Pass) · Verified by Tutor Abdul · June 2017

AQA

Please write clearly in block capitals.

Centre number M F S 0 1 | Candidate number 2 0 2 6

Surname MASTER

Forename(s) FUNCTIONAL SKILLS

Candidate signature Master Functional Skills

OFFICIAL MFS MODEL SOLUTION
Full Marks · 100% Pass
Candidate: Master Functional Skills
Date: 20/10/2026 · Tutor Abdul

Functional Skills Certifi
FUNCTIONAL MATHEM

Level 1

Wednesday 17 May 2017 Morning Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a calculator
- mathematical instruments
- a copy of the Data Book (Examination) (enclosed).



For Examiner's Use	
Question	Mark
1	
2	
3	
4	
TOTAL	

Instructions
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.
- Evidence of checking is specifically assessed in Questions 2(c) and 4(d). These questions are indicated with a †.

Advice
- In all calculations, show clearly how you work out your answer.



MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and healthcare & vocational staff. Everything you need to revise, practice, and pass is at your fingertips.

1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

Ready to Master Your Functional Skills & Pass First Time?

Join thousands of healthcare staff, apprentices, and adult learners across the UK.

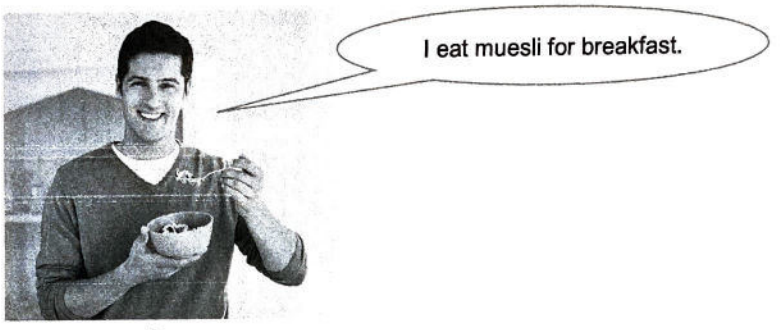
Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

Answer **all** questions in the spaces provided.

1 Muesli



Rory

Rory finds the number of calories and the amount of sugar in different brands of muesli.

	For each serving	
Muesli	**Calories**	**Sugar**
Brand W	222	2 teaspoons
Brand X	170	$2\frac{1}{2}$ teaspoons
Brand Y	219	4 teaspoons
Brand Z	188	$2\frac{1}{2}$ teaspoons

1 (a) For a healthy diet, Rory is allowed 6 teaspoons of sugar each day.
On Monday morning, he has one serving of Brand Z.

How many **more** teaspoons of sugar is he allowed during the day?
Circle your answer.

[1 mark]

2 $3\frac{1}{2}$ $4\frac{1}{2}$ 6

1 (b) Rory has one serving of muesli each day.

This week he has Brand Z on all 7 days.
Next week he will have Brand X on all 7 days.

He says,

"Next week, the muesli I eat will have 150 calories **less** than this week."

Is he correct?

You **must** show your working.

[4 marks]

$188 - 170 = 18$

$18 \times 7 = 126 < 150$

Rory is not correct.

Question 1 continues on the next page

1 (c) For a healthy diet, Rory is allowed 2500 calories each day.
There are 260 calories in his breakfast each day.

He says,
"I have **more than** 10% of my allowed daily calories in my breakfast."

Is he correct?
You **must** show your working.

[3 marks]

10% = 0.1

$0.1 \times 2500 = 250 < 260$

Rory is correct.

1 (d) Lara makes her own muesli.

To make 500 g of muesli she needs

350 g oats	25 g dried apricots
25 g raisins	25 g dried cranberries
40 g nuts	35 g sunflower seeds

She makes these notes.

> 100 g of oats cost 8p
> 100 g of dried apricots cost 72p
> For 500 g of muesli, the other ingredients cost a total of 95p
> A box of similar muesli from a shop costs £2.65 for 500 g

She says,

"For 500 g, my muesli is cheaper than the shop muesli by **more than** £1.20"

Is she correct?

You **must** show your working. **[6 marks]**

100 g costs 8p
8 × 3.5 = 28
350 g costs 28p.
100 g costs 72p.
25g costs 18p
28p + 18p + 95p = £1.41.
£2.65 − £1.41 = £1.24 > £1.20
Lara is correct.

2 **Hot Air Balloon**

There is a **data sheet** for Hot Air Balloon.



Jo

Jo is a private balloon pilot.

She wants to qualify to carry passengers.

2 (a) Jo has already completed eight flights.

These flights lasted a total of 29 hours.

She completes two more flights.

	Day	Start time	Finish time
Flight 1	Friday	1030	1300
Flight 2	Saturday	1600	2000

Jo says,

"I have now completed the hours I need to qualify."

Is Jo correct?

You **must** show your working. **[4 marks]**

1300 - 1030 = 2.5 hours.

2000 - 1600 = 4 hours

29 + 4 + 2.5 = 35.5 hours > 35 hours

Jo is correct.

2 (b) On one of the flights, the balloon reaches a height of 1620 **metres.**

Show that this height is more than 5000 **feet.**

Use the steps on the data sheet.

[3 marks]

$1620 \div 3 = 540$

$540 \times 10 = 5400 > 5000$

†2 (c) Jo rents a balloon with an envelope volume of 3200 m^3

A balloon can lift 1 kilogram for every 4 m^3 of balloon volume.

Work out the maximum number of kilograms this balloon can lift.

[2 marks]

$3200 \div 4 = 800$ kg.

Check your answer.

Show how you have done your check.

[1 mark]

$800 \times 4 = 3200$

MasterFunctionalSkills.co.uk · AQA Maths Level 1 · June 2017 Solutions
MFS Step 7: Accredited AQA Method & Mark Scheme Verified by Tutor Abdul [M1 A1] [100% PASS]

box

2 (d) Jo rents the balloon for a year.

It can carry the pilot and four passengers.

She makes these notes about the flights and costs for the year.

> The flights will start on 1 April and end on 31 October.
> There will be 8 flights each month.
> I will charge each passenger £145 for a ticket.
> My costs for the year will be
> £5700 to rent the balloon
> £2250 for fuel.

Jo says,

"If I sell 4 tickets for every flight, I will make a profit of **more than** £25 000"

Is she correct?

You **must** show your working.

[7 marks]

1 Apr - 31 Oct is 7 months.

Make: $7 \times 8 \times 4 \times £145 = £32480$

$£32480 - £5700 - £2250 = £24530$

$£24530 < £25000$

Jo is not correct.

17

3 Canteen Survey

Betty is the manager of the college canteen.



I want to improve the service we offer.
To help me I will carry out a survey.

Betty

20 students completed Betty's survey form.
Here is one of the completed forms.

Canteen Survey

A Circle the number of days that you used the canteen last week.

0 1 (2) 3 4 5

B Tick a box to show which of these sandwiches you like best.

Egg and Cress	
Bacon, Lettuce and Tomato	✓
Chicken Salad	

C The canteen is thinking of selling sushi.
Tick a box to show what type of sushi you would like best.

Salmon	
Vegetable	✓
Tuna	
None	

3 (a) The mean number of days the 20 students used the canteen that week was 2.5

Betty put some more seats in the canteen.

The following week she asked the same 20 students question **A** again.

Here are the answers.

0	3	5	3	2	2	0	3	3	5
3	4	4	5	3	1	2	2	4	4

Betty says,

"The mean number of days the students used the canteen has increased."

Is she correct?

You **must** show your working.

[4 marks]

$\frac{0+3+5+3+2+2+0+3+3+5+3+4+4+5+3+1+2+2+4+4}{20} = 2.9$

$2.9 > 2.5$

The mean has increased.

box

The table shows information about the answers to question **B**.
It also shows the price of each type of sandwich.

Sandwich	Number of people	Price
Egg and Cress	5	£2
Bacon, Lettuce and Tomato	10	£3
Chicken Salad	5	£3
	Total = 20	

3 (b) Betty makes 80 sandwiches each day.

She uses the survey results to work out how many of each sandwich to make.

Betty works out how much money she could take from selling the sandwiches.

She says,

"If I sell all 80 sandwiches I will take **more than** £200 each day."

Is she correct?

You **must** show your working.

[6 marks]

80 ÷ 20 = 4.

Egg 4 × 5 = 20 20 × £2 = £40

Bacon 4 × 10 = 40 40 × £3 = £120

Chicken 4 × 5 = 20 20 × £3 = £60

£40 + £120 + £60 = £220 > £200.

She is correct.

3 (c) The table shows information about the answers to question **C**.

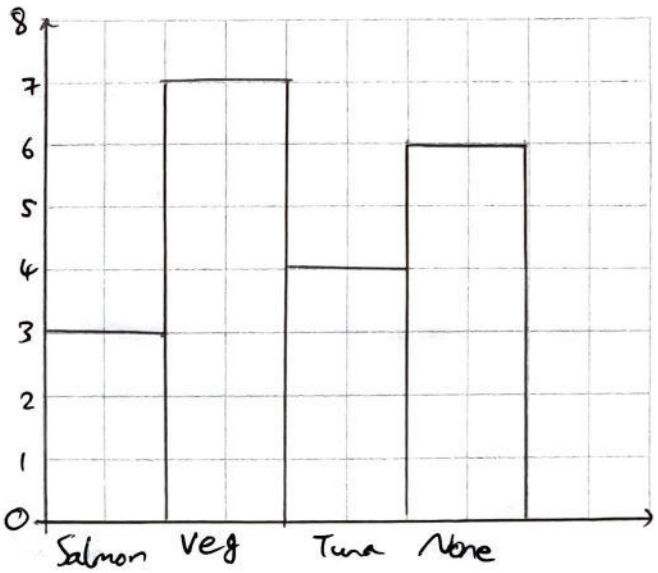
Type of sushi	Number of people
Salmon	3
Vegetable	7
Tuna	4
None	6
	Total = 20

Betty wants to show this information in a diagram.

Draw a pictogram or a bar chart that Betty could use.

Use the space below **or** the grid opposite.

[3 marks]



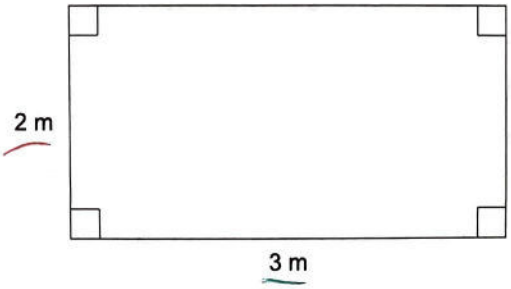
Turn over for the next question

13

4 **Room Makeover**

There is a **data sheet** for Room Makeover.

Here is a sketch of Tom's bedroom floor.



Not drawn accurately

4 (a) The gripper rods in one pack have a total length of 10.6 metres.

Show that one pack has enough gripper rods to go around the edge of the bedroom.

[2 marks]

$2 + 3 + 2 + 3 = 10 < 10.6$

4 (b) The bedroom has height 2.5 metres.
Tom wants a new radiator.

Work out the smallest length of radiator he should buy.

You **must** show your working.

[5 marks]

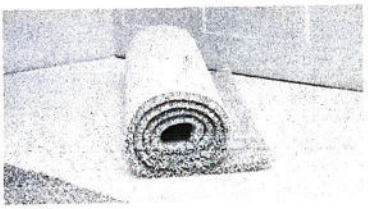
$2 \times 3 \times 2.5 \times 1.41 = 21.15$

Need 6000 mm radiator

MasterFunctionalSkills.co.uk · AQA Maths Level 1 · June 2017 Solutions
MFS Step 14: Accredited AQA Method & Mark Scheme Verified by Tutor Abdul [M1 A1] [100% PASS]

4 (c) Tom is buying a carpet for his bedroom.

He buys this carpet piece, which fits the room exactly.



Carpet piece
Only £95

The shop sells

underlay for £3.99 per square metre
gripper rods for £5.79 per pack.

Tom buys

the carpet piece
enough underlay for the bedroom
one pack of gripper rods.

He says,

"I can buy all these items for **less than** £125"

Is he correct?

You **must** show your working.

[5 marks]

$2 \times 3 = 6m^2$

$6 \times £3.99 = £23.94.$

$£23.94 + £5.79 + £95 =$

$£124.73 < £125.00.$

He is correct.

Turn over for the next question

†4 (d) Tom hires Ali to paint the bedroom.

Ali charges £28 per hour, plus £55 for paint.

He works for 6 hours.

How much does Ali charge in total? **[3 marks]**

£28 × 6 = £168

£168 + £55 = £223.

Check your answer.

Show how you have done your check. **[1 mark]**

(223 - 55) ÷ 6 = 28.

16

END OF QUESTIONS

Copyright Information

For confidentiality purposes, from the November 2015 examination series, acknowledgements of third party copyright material will be published in a separate booklet rather than including them on the examination paper or support materials. This booklet is published after each examination series and is available for free download from www.aqa.org.uk after the live examination series.

Permission to reproduce all copyright material has been applied for. In some cases, efforts to contact copyright-holders may have been unsuccessful and AQA will be happy to rectify any omissions of acknowledgements. If you have any queries please contact the Copyright Team, AQA, Stag Hill House, Guildford, GU2 7XJ.

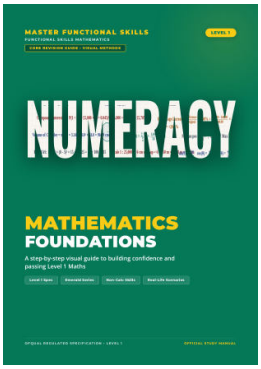
Copyright © 2017 AQA and its licensors. All rights reserved.

MASTER FUNCTIONAL SKILLS STORE

OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

Everything You Need To Revise, Practice & Pass Your Maths Level 1 Exam



Core Curriculum Manual

Maths Level 1
Core Study Guide

- Full official curriculum coverage
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

Spiral Bound Print & PDF

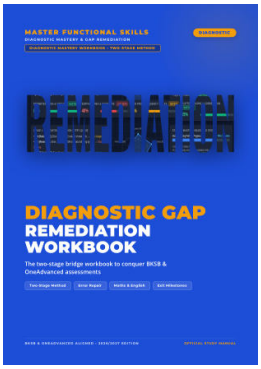


1,000+ Practice Drills

Maths Level 1
Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF



Realistic Mock Papers

Maths Level 1
Official Mock Packs

- Full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF



Pocket Flashcards A6

Maths Level 1
Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by training providers, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk